

CS481 HW4

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Q1

Each Producer puts a value and each Consumer gets a value, it is dependent on the scheduler and the timing of the program since there is only one value in the buffer. Below are the possible outputs.

1. `*val1 = 1, *val2 = 2`
2. `*val1 = 2, *val2 = 1`

Q2

Once again this is based on the scheduler, so if the Consumers run first before the buffers have values, both will go to sleep and wait for the Producers.

Q3

An ordering for a Consumer calling `get` while the other sleeps can be the case where the scheduler runs Consumer1 first while the buffer is empty so it sleeps. Then, calls either of the Producers, then Consumer2, which will run `get` since there is a value in the buffer.

Q4

Using 2 condition variables, `Full` and `Empty`, that signal Consumers to sleep on `Empty` and Producers to sleep on `Full`.

Q5

If we used `If-Else` statements in the code instead of `While-Loops`, The code could break if we wake the Consumers while the buffer is empty causing them to go to sleep. Then if Producer1 runs, we could wake both Consumers causing Consumer1 to get a correct value, emptying the buffer. At the same time, the

buffer is emptied and Consumer2 calls get on the empty buffer which could return an error or the wrong value.